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Transmission line monitoring system testing

Abstract

In some embodiments, optical fiber monitoring may be facilitated. In some embodiments, monitoring of an optical fiber may be performed via a detection system to detect changes in the optical fiber that are indicative of one or more events. During the monitoring, a monitor signal may be transmitted along the optical fiber via a light source, where the monitor signal is received by the detection system as light from the optical fiber. While the detection system continues to receive light from the optical fiber in connection with the transmitting of the monitor signal, a test disturbance event may be caused to occur at a predetermined location along the optical fiber via a device at the predetermined location. In response to a failure of the detection system to detect test-related changes in the optical fiber caused by the test disturbance event, a notification indicating an issue may be generated.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/966,235, filed Oct. 14, 2022, which is a continuation of U.S. patent application Ser. No. 17/089,186, filed Nov. 4, 2020, which is a continuation-in-part of U.S. patent application Ser. No. 17/018,535, filed Sep. 11, 2020, each of which are incorporated herein by reference.

BACKGROUND INFORMATION

(1) Optical fiber is used for many types of monitoring applications, including but not limited to perimeter security, network security, structural monitoring.

(2) Typically the fiber concerned is monitored using a method which includes transmitting from a source of light at a transmit location a monitor signal along the optical fiber, receiving the monitor signal after transmission along the fiber, analyzing the monitor signal after transmission along the fiber to detect changes therein and generating an alarm in response to the detected changes.

(3) In regard to communications networks, the monitor system is responsive to vibration, motion, or handling of the fiber which are indicative of an intrusion attempt on the fiber.

(4) In addition, the invention herein can be used for fence and buried perimeter protection systems where a fiber is mounted on or at the item to be secured so that again the fiber is monitored for

vibration or motion of the fiber caused by attempts to access or penetrate the item concerned.

(5) Yet further, the invention herein can be used for other fibers used for monitoring forces on the fiber caused by strain or other forces that monitor bridge or building integrity. These can include stretching or compression of the fiber. In this case the monitor is not looking for transverse vibration or movement of the fiber from an intrusion attempt or other handling but is instead looking for changes in the character of the fiber caused by the application of the forces to the fiber. Such arrangement can be used in strain gauges, building and bridge monitoring systems and the like.

(6) Additionally, the invention herein can be used to monitor temperature, strain, and pressure using sensors in wells and down-hole applications. Such methods require a looped fiber to accommodate the inability to place equipment in wells or down holes.

(7) Additionally, the invention herein can be used to monitor fibers distributed throughout a so-called 'smart city' type application. In these instances, fibers are distributed to monitor traffic patterns, weather, electrical distribution, and seismic activity.

(8) In all cases the change in the parameter to be measured causes a change in a characteristic in the fiber which can be measured using known techniques.

(9) One method for monitoring a communications network cable is to use fibers that are internal to the protected cable. This so-called "intrinsic monitoring" is shown in U.S. Pat. No. 7,706,641, issued Apr. 27, 2010, the disclosure of which is incorporated herein by reference.

(10) The optical fibers can be monitored using a variety of detection techniques including: Modal metric, where changes in a modal power distribution in a multimode fiber are detected as shown in U.S. Pat. No. 7,092,586, issued Aug. 15, 2006, the disclosure of which is incorporated herein by reference. Attenuation, where simply an attenuation in the monitoring signal received is measured.

(11) Optical Time Domain Reflectometer (OTDR) where reflections or localized attenuations from components of the fiber are detected.

(12) Distributed Sensing (DAS/DSS/DTS):

(13) DAS—Distributed Acoustic Sensing where vibrations and displacements cause localized shifts in the path length of the optical fiber. This is detected by a high precision optical Time Domain Reflectometer (OTDR). This OTDR is often referred to as a Phase-OTDR or ϕ -OTDR, and measures changes in the distance between points of Rayleigh backscatter. DSS—Distributed Strain Sensing—where strain is measured along a fiber due to tensile or compressive displacements, compression, or cracks. Typically measured using Brillouin OTDR, transmitted light and scattered light are mixed as a heterodyne receiver. This Brillouin frequency shift is proportional to strain and temperature in the fiber. DTS—Distributed Temperature Sensor—where temperature is measured along an optical fiber including by use of Raman OTDR. Light propagating down the fiber at two wavelengths cause Stokes and anti-Stokes light. The amplitude of light reflected back to the detector in a similar fashion to Rayleigh Backscattering in a traditional OTDR, is highly dependent on temperature. The ratio of Stokes and anti-Stokes light indicates temperature, while the round trip transit time indicates location. As with DSS, Brillouin OTDR can also be used to measure distributed temperature.

(14) Polarization monitoring, where changes in a polarization in the signal in a single mode fiber are detected as shown in U.S. Pat. No. 7,142,737, issued Nov. 28, 2006, the disclosure of which is incorporated herein by reference.

(15) Active fiber monitoring, where monitoring signal and data signal pass on the same fibers as shown in U.S. Pat. No. 7,092,586, issued Aug. 15, 2006, the disclosure of which is incorporated herein by reference.

(16) Strain monitoring such as strain gauge where a Fiber Bragg Grating, strain gauge or DSS monitors a fiber or mechanical structure, disturbance will be stretching or compression.

(17) Interferometry such as the Mach-Zehnder interferometers used for network and perimeter monitoring. These may be zone based, or locating by use of bidirectional differential time of flight

systems.

(18) Each of these methods of monitoring exploit a specific attribute of the fiber—be it loss, rotation of state of polarization, Rayleigh scattering, or others.

(19) In some cases a single fiber is monitored with typically the transmission at one end and the monitoring at the other or same end. However other arrangements can be used in the present invention including, for example, the loop type network shown for example in U.S. Pat. No. 7,142,737, issued Nov. 28, 2006, the disclosure of which is incorporated herein by reference.

(20) Often these systems are placed into service, but continued viability and availability should be verified. Accordingly, an intelligent, automatic test is needed that does not interfere with or disrupt the actual performance of the monitoring fiber. The test should exercise the entire system, including perturbation detection, alarm generation, response to alarm, and alarm logging. Thus, it is desirable to create a controlled and characterized event on the monitor fiber structure itself to test the entire system and responses.

(21) One existing arrangement for this function is provided by the StopLight (trademark) available from CyberSecure IMS. This test is implemented by passing the monitoring fiber in a secure network through an optical shutter that simply shuts off transmission of the light to the fiber. At the appropriate time, scheduled, spontaneous, or random, the optical shutter opens the circuit, causing the monitoring equipment to register an alarm. This method is accepted in the industry, although a shortcoming is that the monitoring is “blind” during this short period of testing. This adds the vulnerability of an inside threat, cognizant of the blind period, which could act to switch the monitored fibers with a substitute fiber or fiber pair. The system owners would then operate under the belief that the system was secure, although in fact it is no longer monitored.

SUMMARY

(22) According to a first aspect of the invention, there is provided a method for operation of an optical fiber monitoring system comprising: using a detection system to detect changes in an optical fiber caused by one or more events on the fiber by: transmitting from a source of light at a transmit location a monitor signal along the optical fiber; receiving the monitor signal after transmission along the fiber; analyzing the monitor signal after transmission along the fiber to detect changes therein caused by the event to be monitored; and generating an alarm in response to said detected changes which are indicative of an event; generating a waveform which is arranged to trigger an alarm; and periodically optically injecting a test signal corresponding to said waveform into the fiber to test operation of the detection system.

(23) According to a second aspect of the invention, there is provided a method for operation of an optical fiber monitoring system comprising: using a detection system to detect changes in an optical fiber caused by one or more events on the fiber by: transmitting from a source of light at a transmit location a monitor signal along the optical fiber; receiving reflected signals after transmission along the fiber; analyzing the reflected signals after transmission along the fiber to detect an event to be monitored and to detect by timing of the reflected signals a location of the event along the fiber; and periodically optically injecting a test signal into the fiber to test operation of the detection system; wherein the test signal is injected at a time selected to appear as an event at a predetermined location on the fiber.

(24) In one embodiment, the test signal is injected at a time selected relative to the pulse of the monitor signal to appear as a reflection at a predetermined location on the fiber.

(25) In one embodiment, the test signal is injected at a far end of the fiber and wherein there is provided a communication system for actuating timing of the test signal injection.

(26) In one embodiment, the test signal is injected at a remote location of the fiber and wherein there is provided a receiver at the remote location for detecting a time of transmission of the monitor signal.

(27) The test signal can be injected by being backscattered into the fiber to be monitored or direct injected by a laser.

(28) The test signal can be a wave form which can be for example a sine wave or a complex waveform.

(29) According to another feature of the present invention, there is provided a method for verifying operation of an optical fiber monitoring system comprising: using a detection system to detect changes in an optical fiber caused by an event to be monitored by: transmitting from a source of light at a transmit location a monitor signal along the optical fiber; receiving the monitor signal after transmission along the fiber; analyzing the monitor signal after transmission along the fiber to detect changes therein caused by the event to be monitored; and generating an alarm in response to said detected changes which are indicative of an event; and periodically checking proper operation of the optical fiber monitoring system by: providing a fiber disturbance actuator to cause disturbance of a portion of the fiber at a predetermined location thereon where the disturbance is characteristic of an event to be monitored; periodically operating the fiber disturbance actuator; analyzing the monitor signal to detect changes therein caused by said fiber disturbance actuator; and in the event that expected changes in response to said fiber disturbance actuator are not detected, actuating a warning that the intrusion detection system is not properly operating.

(30) The fiber concerned can be used for transmission of data as a single transmission fiber or as part of a network. Alternatively, the fiber may be used for monitoring movement of or damage to the fiber as part of a perimeter or structure monitoring system where the signal is used to detect the movement or vibrations of the fiber as a part of an intrusion event into the area being monitored.

(31) Preferably the fiber disturbance actuator generates a predetermined pattern of movements which can thus be recognized by a signature associated with the pattern.

(32) Preferably the fiber disturbance actuator generates predetermined periodic displacements to the fiber typically at a predetermined frequency. These displacements can be formed in bursts of a series of spaced envelopes each containing predetermined periodic displacements to the fiber.

(33) Preferably the fiber disturbance actuator generates a modulation frequency which is chosen to be out of band with ambient disturbances.

(34) Preferably the fiber disturbance actuator generates a modulation frequency which provides a duty cycle defining the frequency bursts which is chosen to be dissimilar to natural occurrences.

(35) Preferably the method includes analyzing the monitor signal and adjusting a magnitude of disturbance caused by the fiber disturbance actuator in response to the analysis. This will allow the amplitude of the perturbation to be adjusted to be appropriate to the magnitude of the monitor signal so that the amplitude must be high enough to be detected, but not so high as to interfere with the monitor signal processing. The adjustment can be controlled from any of several locations, including over a network from the monitor end of the fiber, or controlled locally such as a local network, manual adjustment, or serial connection.

(36) The amplitude of the test perturbation can be monitored to detect fiber quality as insertion loss changes or the insertion of a fiber optic tap along the fiber.

(37) The arrangement herein thus provides a system for applying a known disturbance or perturbation on a monitor fiber with the intention of being detected by the monitoring system.

(38) This invention excites the fiber under very specific conditions in order to confirm that the fiber, cable, or structure is being monitored as expected.

(39) In a preferred arrangement, the fiber being monitored includes or has connected as a part an input fiber length for isolation from the monitoring device of any connector reflections caused by connection of the fiber disturbance actuator. In many cases this includes also an exit fiber which again acts for isolation purposes. This isolation provided by the inlet and exit fiber portions is particularly important in the systems which rely on reflection techniques where any such reflections can be orders of magnitude greater than the signals to be detected.

(40) This system including the inlet and exit isolation fiber portions is typically connected to a network and is instructed to commence a test. The internal fiber disturbance actuator is engaged in a pre-determined manner, the signal from which is then recognized by the monitoring device,

which in turn registers the test as successful and proves that the fiber being monitored is connected and secure.

(41) In a preferred embodiment, where the monitor equipment is of a type that determines location of the movement rather than simply the existence of the movement, the monitoring system acts to look for a specific location and for a specific signal signature. This will not only ensure the invention is still connected to the monitored fiber, and by applying detection to a very specific location on the fiber it ensures that a substitute fiber is not used to bypass the intended monitored cable.

(42) This system satisfies a number of applications:

(43) Under command from a communicating device such as a data network or serial port, the fiber disturbance actuator can be instructed to activate. When this activation is detected by the monitoring device, satisfaction of periodic health check of the monitoring system is satisfied. This test may be in satisfaction of a formal standard, or as good practice as defined by the user.

(44) The fiber disturbance actuator may have a manual trigger such as a push button, causing a test to initiate. This test might occur immediately or after a predetermined or random time, adding confidence to the test.

(45) The fiber disturbance actuator and the control system operating the actuator can in some cases be configured to initiate tests autonomously; where this might be scheduled or random in occurrence.

(46) The fiber disturbance actuator may be used as a health monitor of the security system. In this embodiment, the control system can initiate a test at a predetermined frequency, verifying the system is functional.

(47) The arrangement can perform periodic connectivity test to confirm the intended fiber is being monitored.

(48) As cited above in U.S. Pat. No. 7,706,641, fiber security products, such as the Interceptor (trademark) product from Network Integrity Systems perform a fiber monitoring function that is zone based. Rather than pinpoint a location, these devices utilize a loop of fiber, and monitor the entire continuous loop as one zone. The arrangement herein is applicable for zone use as well. The fiber disturbance function of the fiber disturbance actuator acts similarly to a location determining product.

(49) In accordance with a further feature of the invention, which can be used with any of the above features, there is provided a method for verifying operation of an optical fiber monitoring system comprising: wherein the optical fiber has a first end and a second end; using a detection system to detect changes in an optical fiber caused by an event to be monitored by: at the first end of the optical fiber transmitting from a source of light a monitor signal along the optical fiber; at the second end returning the monitor signal along the fiber to the first end; at the first end of the optical fiber receiving the monitor signal after transmission along the fiber; analyzing the monitor signal after transmission along the fiber to detect changes therein caused by the event to be monitored; and periodically checking proper operation of the optical fiber monitoring system by: providing an optical shutter at the second end of the fiber; periodically operating the optical shutter to temporarily terminate transmission of light along the fiber; and analyzing the monitor signal to detect a termination therein caused by said optical shutter.

(50) In accordance with a further feature of the invention, which can be used with any of the above features, there is provided a method for verifying operation of an optical fiber monitoring system comprising: using a detection system to detect changes in an optical fiber caused by an event to be monitored by: transmitting from a source of light at a transmit location a monitor signal along the optical fiber; receiving the monitor signal after transmission along the fiber; and analyzing the monitor signal after transmission along the fiber to detect changes therein caused by the event to be monitored; and periodically checking proper operation of the optical fiber monitoring system by: at a predetermined location on the fiber periodically operating an actuator to cause a disturbance

which changes the monitor signal in the fiber; analyzing the monitor signal to detect changes therein caused by said disturbance; and in the event that expected changes in response to said disturbance are not detected, actuating a warning that the intrusion detection system is not properly operating; wherein an instruction signal is communicated along the fiber to the actuator to effect said operating of the actuator and there is provided a coupler on the fiber to extract the instruction signal from other signals in the fiber.

(51) In accordance with a further feature of the invention, which can be used with any of the above features, there is provided a method for verifying operation of an optical fiber monitoring system: wherein the system comprises a first fiber and a second fiber; each of the first and second fibers extending from a transmit location to a remote location and returning along a continuous optical path to the transmit location; each of the first and second fibers defining first and second ends at the transmit location thus defining outward and return portions of the fiber; the method comprising: using a detection system to detect changes in an optical fiber caused by an event to be monitored by: transmitting from at least one source of light at the transmit location into the first and second ends of each of the first and second fibers monitor signals so as to travel along the optical fiber; receiving the monitor signals after transmission along the fibers; analyzing the monitor signals after transmission along the fiber to detect changes therein caused by disturbances on the fiber; and periodically checking proper operation of the optical fiber monitoring system by: at a predetermined location on the first and second fibers periodically operating an actuator to cause a disturbance which changes the monitor signal in the fiber; analyzing the monitor signal to detect changes therein caused by disturbance; and in the event that expected changes in response to said disturbance are not detected, actuating a warning that the intrusion detection system is not properly operating.

(52) In accordance with a further feature of the invention, which can be used with any of the above features, there is provided a method for monitoring an optical fiber comprising: transmitting from a source of light at a transmit location a monitor signal along the optical fiber; receiving the monitor signal after transmission along the fiber; analyzing the monitor signal after transmission along the fiber to detect changes therein caused by disturbances on the fiber; and periodically generating disturbances in the fiber by generating longitudinal forces in the fiber at one location along the length relative to another location along the length so as to cause changes in length of the fiber between the first and second locations.

(53) In one specific arrangement, the disturbances can be generated by a longitudinally moving anchor located between two stationary anchors.

(54) In one specific arrangement, the disturbances can be generated by a by wrapping the fiber around a support to form two adjacent lengths and commonly generating said longitudinal forces in said adjacent lengths.

(55) In one embodiment for causing the longitudinal forces, the support comprises a stationary anchor around which the fiber is wrapped and there is provided a longitudinally moving anchor movable towards and away from the stationary anchor. As detection systems monitor interference within the fiber, and therefor may exhibit points of constructive and destructive interference (nulls), it can be desirable to excite more than a single point along the fiber. In FIG. 16, the fiber enters the apparatus, and is fixed to the stationary anchor. The fiber then proceeds to the floating support to which it is fixed, then around the dowel which reverses direction while maintaining proper minimum bend radius of the fiber, and again fixed to the floating support and stationary anchor. By displacing the floating support by the disturbance actuator, longitudinal strain is placed in 4 locations of the fiber—that between the anchor, support, and dowel; in both directions.

(56) In another embodiment, the longitudinal forces in the fiber at are generated by wrapping the fiber around a body having two parts of the body separated by a slit and moving the two parts toward and away from one another. The cylindrical shell supporting the wrapped fiber is stably mounted at the fixed anchors and can be moved at high frequency with low forces at the actuator at

the edge of the slot. In this way the required signal can be applied to the fiber at the required amplitude and frequencies without difficulty and effectively. In this implementation the actuator may be mounted to the free edge of the cylinder at the slit, causing the free edge to move under the influence of the vibrating mass of the actuator. Alternately, the body of the actuator may be fastened to the same support base as the stationary supports and the portion of the actuator that is displaced may be fastened to the free edge, causing vibration without influence of stiffness of the edge, mass of the actuator, or resonances thereof. This vibration stretches and releases tension on the fiber(s), thereby modulating the longitudinal strain.

(57) In accordance with a further feature of the invention, which can be used with any of the above features, there is provided an apparatus for verifying operation of an optical fiber monitoring system comprising: an optical fiber system including a fiber to be monitored; a common housing mounted at an end of the fiber to be monitored; a monitoring system to detect changes in the optical fiber caused by an event on the fiber comprising: a source of light arranged to transmit a monitor signal along the fiber; a transducer arranged to receive the monitor signal after transmission along the fiber; and a monitoring processor arranged for analyzing the monitor signal after transmission along the fiber to detect changes therein caused by the event to be monitored; and a test device for periodically checking proper operation of the monitoring system comprising: a fiber disturbance actuator to cause disturbance of a portion of the fiber at a predetermined location thereon where the disturbance is characteristic of an event to be detected; and a test processor periodically operating the fiber disturbance actuator and for detecting a response of the monitoring processor caused by changes in the monitor signal caused by said fiber disturbance actuator; wherein at least the monitoring processor and the test processor are mounted in the common housing.

(58) In one embodiment, the monitoring processor and the test processor can be parts of a common processor or, in an alternative embodiment, the monitoring processor and the test processor are separate processors where the monitoring processor communicates to the test processor data related to the changes in the monitor signal caused by said fiber disturbance actuator.

(59) In accordance with a further feature of the invention, which can be used with any of the above features, there is provided a method for operation of an optical fiber monitoring system comprising: using a detection system to detect changes in an optical fiber caused by one or more events on the fiber by: transmitting from a source of light at a transmit location a monitor signal along the optical fiber; receiving the monitor signal after transmission along the fiber; analyzing the monitor signal after transmission along the fiber to detect changes therein caused by the event to be monitored; and generating an alarm in response to said detected changes which are indicative of an event; generating a waveform which is arranged to trigger an alarm; and periodically optically injecting a test signal corresponding to said waveform into the fiber to test operation of the detection system.

(60) In one embodiment, the test signal is injected into an added length of optical fiber which is connected to the fiber to be monitored through a device, such as a coupler, splitter, circulator, whereby the test signal is backscattered into the fiber to be monitored. In a monitor system that detects Rayleigh backscatter, it is necessary to present an appropriate signal for detection. In this embodiment, an additional length of optical fiber is added, and that fiber is excited by a laser, causing Rayleigh backscatter to be launched into the monitored fiber and return to the monitoring system.

(61) In this arrangement preferably the added length of optical fiber is terminated in a low reflectance manner.

(62) In this arrangement preferably the added length of optical fiber is terminated using a wavelength specific reflection, such as a Fiber Bragg Grating, which acts to return just the added test signal.

(63) In one embodiment, the test signal is injected from a laser into the fiber at or adjacent a far end of the fiber.

(64) In this arrangement, an amplitude of the test signal can be lowered to correspond to a

backscattered signal. For example, the test signal can be reduced in amplitude by in-line attenuators. In this arrangement, angled connectors and low-reflectance optics can be used to control any back reflection from the detection system.

(65) In one embodiment, the laser at the transmit section of the monitoring device generating the monitor signal is periodically disabled while continuing to monitor with receiver. This allows the monitor system receiver to detect an injected signal in the absence of a monitor signal. It allows more sophisticated messaging that is not buried in monitor signal.

(66) In one embodiment, the transmit signal is sent during off times that are normal to a pulsed laser device. This does a similar effect to the one immediately above with the exception that it ties the timing to the pulses of the monitor system which translates into a distance location. The monitor system sends a pulse and interrogates the reflections coming back as a function of time which equates to distance. Tying an injected signal to that time base shows it at the far end of the fiber.

(67) In one embodiment, the test signal is injected from a laser into the fiber at a position spaced from a far end of the fiber and the signal is transmitted along the fiber by reflection from the far end. In this arrangement preferably a low internal reflectance variable attenuator, optical switch, or both are placed inline with a high reflectance termination. Preferably the termination is an unterminated or air-gap connector. For example, the fiber is terminated with a reflection such as a reflective deposition on a connector face, such as gold.

(68) In one embodiment, the reflection is periodically turned on or off, creating an effective end of line signal that provides many of the features set forth above.

(69) In one embodiment, a variable attenuator is modulated such as with a sine wave, causing the end reflection to vary at a pre-determined rate which adds an additional layer of security as the monitoring device will watch for that frequency at that precise location.

(70) In one embodiment, as explained herein, the test signal is a waveform which is representative of one event of said events. For example, the waveform is generated by recording an actual event.

(71) In one preferred arrangement, the monitor signal is analyzed by the DAS system described above which requires very narrow spectrum lasers of tightly controlled wavelength and a laser used to inject the test signal is controlled and arranged to provide a test signal which meets the requirement of the DAS system.

(72) In one preferred arrangement, the analysis system is modified based on the received test signal caused by said injected signal so as to tune the analysis.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram showing a system of the present invention installed in a fiber to be monitored by inlet and exit isolation fibers.

(2) FIG. 2 is a block diagram showing a system of the present invention installed as a far end connection in a data transmission fiber to be monitored using inlet and exit isolation fibers.

(3) FIG. 3 is a block diagram showing a system of the present invention installed as a near end connection in a data transmission fiber to be monitored using inlet and exit isolation fibers.

(4) FIG. 4 is a block diagram showing a system of the present invention installed as a near end implementation using a separate dedicated monitoring fiber in a data transmission cable.

(5) FIG. 5 is a block diagram showing a system of the present invention installed as a far end implementation using a separate dedicated monitoring fiber in a data transmission cable.

(6) FIGS. 6 and 7 show sample bursts of cyclical disturbances arranged in envelopes which provide a specified signature of period and frequency which allows the monitoring device to identify the applied disturbances from the fiber actuator.

- (7) FIG. 8 shows schematically one example of a fiber disturbance actuator using a motor for physically moving one part of the fiber relative to another.
- (8) FIGS. 9A and 9B show schematically in side and end view respectively one example of a fiber disturbance actuator using an electromagnetic field to physically move a fiber portion which is coated with a material responsive to the field.
- (9) FIG. 10 shows schematically one example of a fiber disturbance actuator which uses a motor to bend one portion of the fiber relative to another.
- (10) FIG. 11 is a schematic illustration of the operating components which form the elements of the invention for attachment to the fiber to be monitored.
- (11) FIG. 12 is a schematic illustration of the operating components which form the elements of the invention for attachment to the fiber to be monitored where the signal in the fiber is disturbed by an optical shutter at the far end of the fiber.
- (12) FIG. 13 is a schematic illustration of the operating components which form the elements of the invention for attachment to the fiber to be monitored where the signal in the fiber is generated as a waveform related to or corresponding to a waveform detected from an actual event.
- (13) FIG. 14 is a schematic illustration of the operating components which form the elements of the invention for attachment to the fiber to be monitored where the actuator which applies the test signal to the fiber is controlled by a communication system along the same fiber which uses a coupler to extract the control signal.
- (14) FIG. 15 is a schematic illustration of the operating components which form the elements of the invention for attachment to the fiber to be monitored where the fiber optic system includes two fibers and the method includes transmitting from at least one source of light at the transmit location into the first and second ends of each of the first and second fibers monitor signals so as to travel along the optical fiber.
- (15) FIG. 16 is a schematic illustration of the operating components which form the elements of the invention for attachment to the fiber to be monitored where the fiber is disturbed by a first arrangement for applying longitudinal forces on the fiber.
- (16) FIG. 17 is a schematic illustration of the operating components which form the elements of the invention for attachment to the fiber to be monitored where the fiber is disturbed by a second arrangement for applying longitudinal forces on the fiber.
- (17) FIG. 18 is a side elevational view of the operating components which form the elements of the invention for attachment to the fiber to be monitored where the fiber is disturbed by a third arrangement for applying longitudinal forces on the fiber.
- (18) FIG. 19 is a top plan view of the arrangement of FIG. 18.
- (19) FIG. 20 is a schematic illustration of the operating components which form the elements of the invention for attachment to the fiber to be monitored where the signal in the fiber is applied by optically injecting the signal into the fiber using a first arrangement where the signal is injected using an optical coupler.
- (20) FIG. 21 is a schematic illustration of the operating components which form the elements of the invention for attachment to the fiber to be monitored where the signal in the fiber is applied by optically injecting the signal into the fiber using a second arrangement where the signal is directly injected and inline attenuation components are used to control an amplitude of the required signal.
- (21) FIG. 22 is a schematic illustration of the operating components which form the elements of the invention for attachment to the fiber to be monitored where the signal in the fiber is applied by optically injecting the signal into the fiber using a third arrangement where an optical shutter is used to control the injected signal.
- (22) FIG. 23 is a schematic illustration of the operating components which form the elements of the invention for attachment to the fiber to be monitored where the signal in the fiber is applied by optically injecting the signal into the fiber using a third arrangement where an optical shutter, variable attenuator, or combination of the two is used to control the injected signal.

(23) FIG. **24** is a schematic illustration of the operating components which form the elements of the invention for attachment to the fiber to be monitored where the signal in the fiber is applied by optically injecting the signal into the fiber using an optical coupler which combines the monitor signal with the injected signal. This holds true for closed loop systems which are non-locating in regard to the location of the event as well as systems which act to locate the event along the fiber.

(24) FIG. **25** is a schematic illustration of the operating components which form the elements of the invention for attachment to the fiber to be monitored in a location sensing system where the test signal in the fiber is applied by optically injecting the signal into the fiber at a remote location and timed relative to a monitor pulse to simulate a particular location on the fiber.

DETAILED DESCRIPTION

(25) As shown in FIG. **1** there is provided a fiber **10** including connector **10A** which provides an input from a monitoring device such as a fiber optic security system. An isolating fiber portion **11** including inlet and exit portions **11A**, **11B** is connected into the fiber **10** and acts to displace the main monitored section of the fiber from the monitoring device. A fiber disturbance generator **12** is mounted in the isolating fiber portion and is controlled by a controller **14**. This causes an input disturbance of a predetermined signature and magnitude, as discussed hereinafter, on the portion of the fiber **10** which is being monitored by the monitoring device.

(26) The controller **14**, which can be actuated by a switch **15** or by a communications interface **16**, from the monitoring system acts for causing the fiber disturbance generator to generate the characteristic disturbance signature with the required magnitude of disturbance.

(27) The communications interface acts to allow the monitor system to initiate a test. The mechanical switch can be used by a system operating person for generating an autonomous test.

(28) In FIG. **3** an installation is provided of the fiber disturbance actuator system including the actuator and the isolation fiber portions at the near end of the fiber is shown. This consists of the system monitor, such as a Distributed Acoustic Sensor interrogator as discussed above. This detects the characteristic disturbance signature introduced by the fiber disturbance generator. As this type of monitoring system is very sensitive to reflections, the isolating fiber isolates the fiber disturbance generator from the reflections caused by the connection to the monitor. Also, the isolation fiber portions act to allow recovery from any deadzone effect of the monitor. For example, any monitor utilizing optical time domain reflectometry (OTDR) exhibits a “deadzone” at the front panel, causing it to be blind to events immediately after the connection. The internal isolating fiber eliminates that issue since in effect the dead zone is moved into the isolation fiber area.

(29) The output isolating fiber portion **11B**, like the input isolating fiber portion **11A**, allows for isolation from the front connector. It also allows the system to be connected bidirectionally. The output of this output fiber portion is connected to the fiber **10** to be protected. The controller section causes the fiber disturbance generator **12** to generate the required signature and amplitude for detection.

(30) The optional external switch **15** can be activated to cause the controller to cause a test to be performed under control of the system operating personnel, that is, without prior instruction from other equipment in the control system.

(31) FIG. **2** shows a version of the invention which is similar to the FIG. **3** with the differentiation that it is intended to be installed at the far end of the fiber, or at any location remote from the near end where power is available. The addition of a communication device **17** connected to a communication medium such as the shown fiber is used to send instructions such as a request for test to the controller and a feedback from the monitor system concerning the amplitude of the disturbances. This has the added benefit of testing the far end of the cable or at a location remote from the near end, which enhances assurance of security.

(32) FIG. **4** illustrates the monitor fiber which intrinsically protects all the fibers in the cable by utilizing a spare fiber in the cable.

(33) FIG. **5** illustrates an embodiment that utilizes a pair of available fibers in a cable: one for

monitoring and one for communications between the monitoring device and the near end equipment. This is a preferred embodiment where a system can be installed on spare fibers in a cable adding connectivity assurance as well as health tests and periodic testing to the cable monitor. (34) FIGS. 6 and 7 show typical patterns P1, P2 and P3 of fiber displacement where the fiber disturbance actuator generates a series of spaced envelopes of signal bursts each containing predetermined periodic displacements to the fiber. In FIG. 7, the frequency of the bursts in the pattern P2 is increased relative to that of FIG. 6 and the period of the bursts is decreased. It will be appreciated that various patterns can be generated to create a signature pattern to be detected by the monitoring system. Preferably the frequency of the pulses is chosen to be out of band with ambient disturbances. Preferably the modulation frequency provides a duty cycle defining the frequency bursts which is chosen to be dissimilar to natural occurrences which can be expected from machine learning so that the signature pattern can be readily determined during the test process. This can include repetitive duty cycle such as 50:50. The duty cycle defining the frequency bursts may be a complex keyed code for security. This can be changed periodically, randomly, on a scheduled basis, or triggered by a coded or uncoded message or by detectable amplitude changes to the envelope as shown in pattern P3.

(35) In FIG. 7 the last pulses in pattern P3 are shown at reduced amplitude which is determined by analysis of the received monitoring signals so that the amplitude matches a requirement to provide signals which are large enough to be identified and not so large that they interfere with the normal monitoring process by having changes which are beyond those which are expected and are measurable.

(36) When used for near end applications, system can be implemented with only the first isolation fiber, which is provided between the monitoring device and actuator device.

(37) When used for the far end application, system is implemented with both input and output fibers as a method for isolating the actuator signal from reflections.

(38) The system with input and output fiber spools can be used either near end or far end, and is bidirectional in that it can be optically connected in either direction.

(39) It will be appreciated that the type of disturbance used by the system actuator is selected to match the technology of the monitoring device; thus for example:

(40) In a modal metric detection system, the actuator typically uses a disturbance device which acts to bend or physical move a portion of the fiber.

(41) In an attenuation detection system, the actuator typically uses a disturbance device which acts to bend the fiber.

(42) In an Optical Time Domain Reflectometer detection system, the actuator typically uses a disturbance device which acts to bend the fiber.

(43) In a Distributed sensing (DAS/DSS/DTS) detection system, the actuator typically uses a disturbance device which acts to create changes in the monitor signal which can be detected by this type of monitoring system.

(44) In a Distributed Acoustic Sensing (DAS) detection system, the actuator typically uses a disturbance device which acts to move the fiber in a shaking or vibrating action, or by inducing strain such as by stretching the fiber.

(45) In a Strain monitoring system such as strain gauge detection system or a Distributed Strain Sensing system (DSS), the actuator 12 shown in FIG. 1 typically uses a disturbance device which acts to stretch or compress the fiber.

(46) In a Distributed Temperature Sensing (DTS) detection system, the actuator 12 in FIG. 1 typically uses a disturbance device which acts to heat or cool the fiber.

(47) In a Polarization detection system, the actuator typically uses a disturbance device which acts to bend or shake the fiber, or mechanically rotate the state of polarization such as by moving, shaking, or vibrating paddles which introduce birefringence by changing stress on the fiber by way of bending or rotating.

- (48) In an Interferometry detection system, the actuator typically uses a disturbance device which acts to bend or shake the fiber.
- (49) As an alternative an active area of actuator can contain a fiber Bragg grating, in which the actuator acts to heat, bend, or stretch the grating.
- (50) The communications interface may contain a dual wavelength/single fiber ethernet connection. When used with a single fiber monitor, a two fiber solution will protect a cable. When installing optical cables, it is common practice to install cables with more fiber than is needed for the immediate or foreseeable future. As the bulk of the price of an installation is labor, and the price difference when upgrading the fiber count is incremental, unused (called "dark") fiber are often available. As networks most often utilize 2 fibers each, one for transmit and one for receive, and as fiber count in cables is typically an even number, often a multiple of 6 or 12, there are often pairs of optical fibers that are available for use. Single fiber monitoring systems, such as DAS, will utilize one fiber in a pair while leaving the other available for communication. Single fiber communications standards, such as Ethernet, provide full duplex communication over a single fiber by transmitting one wavelength, such as 1310 nm, in one direction, and another wavelength such as 1550 nm in the opposite direction. Combining the single fiber monitoring with the single fiber communication provides a complete instance of this invention on a pair of fibers. Single fiber communications solutions can also be achieved by techniques such as differing modulation frequencies, states of polarizations, time division multiplexing, and others.
- (51) Similarly, the communications might be realized by use of multiple fibers, such as two. Monitoring and communications functions can be performed over the same fiber pair(s) by use of wavelength division multiplexing, time division multiplexing, or other multiplexing schemes.
- (52) Zone based non-locating monitoring methods are often implemented over two fibers, or a fiber loop. Use of multiplexing methods such as wavelength division multiplexing, time division multiplexing, or other multiplexing schemes can be utilized to share the fibers between the monitoring and the communication systems.
- (53) CW detecting monitoring devices, such as zone type network or perimeter protection devices can detect the frequency with a frequency detecting method.
- (54) This method can be a hardware phase locked loop within the receiver circuitry of the monitoring equipment. In this embodiment, a disturbance of known frequency is generated while frequency detection equipment such as a phase locked loop is used for detection, which registers a successful test.
- (55) This method can be a software phase locked loop. Similar to the hardware phase locked loop within the receiver circuitry of the monitoring equipment, a detection algorithm is used within the signal processing software. In this embodiment, a disturbance of known frequency is generated while frequency detection algorithm emulating a phase locked loop is used for detection, which registers a successful test.
- (56) This method can be bandpass filters. Similar to the hardware phase locked loop within the receiver circuitry of the monitoring equipment, a hardware or software frequency filter can suppress all frequencies except that of the disturbance generator. In this embodiment, a disturbance of known frequency is generated while detection equipment detects the signal passed through the bandpass filter, which registers a successful test.
- (57) This method can be Fourier Transforms. The spectra of a received signal can be inspected for the presence of the disturbance frequency. In this embodiment, a disturbance of known frequency is generated while detection equipment detects the signal as a spectral spike of sufficient amplitude, which registers a successful test.
- (58) This method can be correlation, including Wavelet Transforms. The spectra of a received signal can be inspected for the presence of the disturbance frequency. In this embodiment, a disturbance consisting of a pulse, chirp, wavelet, or other finite signal of known composition is generated while detection equipment detects the signal and applies correlation or Wavelet

Transform to detect the presence of the disturbance, which registers a successful test.

(59) Zone based products may multiplex communication signal and monitor signal on the same fiber pair by using wavelength division multiplexers. Zone based systems may be configured for single mode or multimode fiber.

(60) A system as described, but used for zone-based non-location determining systems, can omit the internal isolations fiber portions as reflections and dead zones are not of concern.

(61) For vibration sensing monitoring sensors, the active device disturbing the fiber within the disturbance generator may be of several technologies, including but not limited to:

(62) Electromagnetic actuators which cause displacement by energizing a coil or other electromagnetic device. This might include attaching a fiber to a moving portion of a voice coil.

(63) As Electromagnetic actuators which cause displacement by coating by depositing or other technique a sensitive material to the fiber and placing it within the field with no other moving parts.

(64) The fiber disturbance device **12** shown in FIG. **1** can use a Piezo-electric actuator attached to the fiber and causing the vibration.

(65) The fiber disturbance device **12** shown in FIG. **1** can use a Haptic actuator such as the rotating motor type.

(66) Mechanical Such as Rotating Cam or Sawtooth.

(67) The fiber disturbance device **12** shown in FIG. **1** can use an arrangement where the fiber is displaced by placing it between stators within an electrostatic field, and varying the field to displace the fiber

(68) Variable optical attenuator: perturbation is a variation in optical signal amplitude.

(69) Variable polarization controller: perturbation is variation in optical signal polarization.

(70) Variable optical mode mixing: use a mode mixer to change modal fill of optical signal in MMF fiber, causing a perceived perturbation in our products.

(71) In the described system, a fiber Bragg grating may be used within the disturbance generator. When perturbed, detectable wavelength shifts are detected.

(72) In systems that determine distance or location, the detection signature should occur at a predetermined location, representative of the installation. This precise location thwarts attempts to spoof the system by bypassing with a separate fiber. This attempted spoofing might, at the fiber patch panel or other convenient locations, replace the connections to the monitored cable with a fiber. Requiring a precise fiber length and event location significantly eliminates that ability.

(73) The disturbance repetition rate, frequency, or combination of these and other parameters may be arranged to be representative of a unique key.

(74) The system may be placed at the beginning, end, or any location along the path of the fiber where electrical power is available.

(75) The device is preferably arranged to produce a heartbeat at a configurable, identifiable frequency and cadence that the monitoring interrogator can reliably interpret as a unique event.

(76) When operating in periodic mode the device can be configured with a cryptographic key to communicate time based one-time passwords (TOTP) during tests. This can be used to prove the identity of the test device to the sensing device. The password can be encoded by the content of the test signal. The password can be encoded by the timing of the test. In this way, devices operating in the above time based one-time password mode can be used not just for testing the sensing system, but for providing evidence that that sensing cable has not been bypassed. Devices operating in periodic mode only without network access may be battery powered.

(77) The disturbance Generator will have an adjustable magnitude which will allow perturbation to be adjusted to be appropriate to the magnitude of the monitor signal. The amplitude must be high enough to be detected, but not so high as to interfere with signal processing. This may be controlled from any of several locations, including over a network from the monitor end of the fiber, or controlled locally such as a local network, manual adjustment, or serial connection.

(78) FIG. **8** shows schematically one example of a fiber disturbance actuator using a motor **30**

operating a rotating cam **31** for moving one part of the fiber relative to another part which is held fixed by an anchor **32**.

(79) FIGS. **9A** and **9B** show schematically one example of a fiber disturbance actuator using an electromagnetic field generator **40** to physically move a fiber portion **41** which is coated with a material **42** responsive to the field.

(80) FIG. **10** shows schematically one example of a fiber disturbance actuator which uses a motor **50** to bend one portion of the fiber relative to another.

(81) FIG. **11** shows schematically the components herein. Switch **18** represents a mechanical input device which, when actuated, initiates a test or a sequence that leads to the performance of a test. This can also represent other types of interfaces such as contact closures and/or other interfaces to allow the test device to send notifications when certain activities happen in and around the device; i.e., door closures or equipment cabinet openings. Signatures generated by the test device can communicate notifications of inputs at the test device, and report to the monitoring device.

(82) As set forth above, various disturbance mechanisms are disclosed for causing a test to be performed using various arrangement for disturbance of the fiber.

(83) FIG. **1** describes a first embodiment which uses an actuator **12** for disturbing the fiber by mechanical movement. In FIG. **12** is shown an alternative arrangement using many of the same components of FIG. **1** where there is disclosed the concept where the disturbance is carried out by an optical shutter or switch **12A** which is actuated to break the path in the fiber. This provides compliance to Federal regulations for self-test, but rather than perform them directly after the monitor device, the test is performed at the far end of the fiber, certifying that the fiber being monitored is what is expected.

(84) As set forth above, the shutter **12A** is operated to provide a disturbance of the fiber in a recognizable pattern to assure that the expected fiber is being monitored.

(85) The method of testing therefore includes the steps of periodically checking proper operation of the optical fiber monitoring system by providing the optical shutter **12A** at the second end of the fiber remote from the monitoring system **10A**, periodically operating the optical shutter by the controller **14** under operation from the communication system **16** to temporarily terminate transmission of light along the fiber **10** and analyzing the monitor signal at the monitoring system **10A** to detect a termination in the monitor signal caused by the operation of the optical shutter **12A**.

(86) Turning now to FIG. **13** there is shown a further modification where the testing system is improved by storing complex waveforms and recordings **14A** in the controller **14** of the test device. These waveforms are representative of an event such as expected intrusions such as a fence climb or fence cut. That is in this method a simulation of an actual event is carried out at event **10C** at a location along the fiber where events are expected to occur and the effect of that simulated event is detected by the signal analysis components of the monitoring system **10A** so as to generate a waveform. Different events can be simulated at different locations to create a library of waveforms each representative of a respective event. Rather than recordings actual events, waveforms can be generated by modelling or simulation to create the library. Or the library can contain actual events and simulated events so as to be able to carry out a series of tests on the system.

(87) Thus, this system includes the steps of generating a waveform **14A** which is representative of one event of the library of events, at a predetermined location on the fiber periodically operating the actuator **12** to cause a disturbance on the fiber which changes the monitor signal in the fiber; where the actuator **12** is operated by the control system **14** based on the waveform **14A**.

(88) This is valuable for at least 2 reasons:

(89) The test device playing the recording simplifies a portion of the tuning during the initial installation. A library of intrusions or signals is played while the installation personnel adjusted the monitoring device without the need for a second person performing the intrusions. This allows for smaller work crews for the calibration process.

(90) The test device can play back a recording of an intrusion as part of the routine testing of the

monitoring system. Rather than just verify the fiber is being monitored, this allows periodic verification that the monitoring system is performing correctly by detecting representative intrusions. These are not reported as actual intrusions at the monitoring system **10A** as the location and timing of the test signal is known to the test device. This functionality adds a layer of protection in that, it not only assures the proper fiber is being monitored, it also assures that they detection calibration has not been altered or desensitized since the set up calibration.

(91) Turning now to FIG. **14** there is shown an arrangement in which the communication to the controller **14** of the actuator **12** is provided by adding a coupler **16B** or other device within or attached to the test device. In this way a portion of the signal fed to an optical detector **16C** allows the near end monitoring system **10A** to send commands encoded in the test laser pulses through the same fiber **10**. This eliminates the need for additional network connectivity to the far end which uses a second fiber as shown in FIG. **1**. Thus, in this arrangement an instruction signal is communicated along the same fiber **10** to the actuator **12** to effect the operating of the actuator and there is provided a coupler **16B** on the fiber to extract the instruction signal by the detector **16C** from other signals in the fiber to communicate with the actuator **12** through the controller **14**.

(92) Turning now to FIG. **15** is shown a pair of monitored fibers, each with a monitor system and with a termination in the same location. In a single fiber locating system, it is common practice to monitor two fibers, often within the same cable, with 2 channels of equipment, and the signals propagating in opposing directions: one traverses the cable clockwise, the other counter-clockwise. This is done for cut protection, if the cable is cut or damaged, one channel monitors CW up to the break, the other channel CCW to the break.

(93) This performs a full fiber monitoring scenario even in the instance of a fully severed fiber cable. In the implementation shown above, monitoring the end of the return cable performs the monitoring function on both cables as though it were a far end implementation. Furthermore, using a single test device eliminates some cost by two disturbance actuators disturbing both fibers together, or separate actuators for independent control. Other circuit components can also be shared including the power supply, controller, etc.

(94) A secondary implementation of—FIG. **15** shows a 2-fiber looped back monitoring system. Rather than open the optical test circuit at the near end, this acts to break the connection at the far end, assuring the correct fiber is being monitored. This holds true for closed loop systems which are non-locating in regard to the location of the event as well as systems which act to locate the event along the fiber.

(95) Thus, the system includes a first fiber **101** and a second fiber **102** where each of the first and second fibers extends from the transmit location at the monitoring system **10A** to a remote location **103**, **104** and returning along a continuous optical path to the transmit location. Each of the first and second fibers defines first and second ends at the transmit location **10A** thus defining outward **104**, **105** and return portions **106**, **106** of the fiber. In this case the actuator **12** is located at the termination end of the fibers and arranged to operate on both fibers. The actuator can act on one leg or both legs of the fibers.

(96) Adding the disturbance actuator **12** at the termination end of the multiple fiber loops allows functionality as follows:

(97) Additional implementation of the single disturbance actuator design is the ability to simplify the test device design by the use of multistrand fiber, such as ribbon fiber or bifilar fiber, with the multiple fibers passing through the actuator. The entire test device is the same except for the number of connectors and the internal multi-strand fiber.

(98) For near end implementations, the monitoring device and the test device could be built into the same mechanical chassis.

(99) Additionally, in two-fiber closed loop monitoring systems, this can perform the routine test as required by some federal guidelines for assurance of the fiber monitoring system. By either shared actuators or just shared control electronics, a single rack space device could monitor multiple

channels, for example matching an industry standard 4-channel monitoring device.

(100) Turning now to FIGS. **16** to **19** there are shown arrangements in which an actuator for periodically generating disturbances in the fiber includes components generating longitudinal forces in the fiber at one location along the length relative to another location along the length so as to cause changes in length of the fiber between the first and second locations. That is, as shown in FIG. **17**, the disturbances are generated by a longitudinally moving anchor **60** located between two stationary anchors **61** and **62**. In this arrangement an electromagnetic actuator **12** provides movement of the anchor **60** longitudinally of the fiber so that the portions of the fiber between the stationary anchors and the floating support are stretched and released alternately to generate longitudinal strain in the fiber. This has been found to generate the necessary changes in fiber structure to introduce the signals.

(101) In FIG. **16** the disturbances are generated by wrapping the fiber around a cylindrical support body **65** to form two adjacent lengths **10X** and **10Y** and commonly generating the longitudinal forces in the adjacent lengths. That is the support comprises a stationary wrapped anchor **65** around which the fiber is wrapped and there is provided a longitudinally moving anchor **60** movable towards and away from the wrapped anchor as well as from stationary anchor **61**.

(102) In FIGS. **18** and **19** is shown another arrangement for stretching the fiber where the longitudinal forces in the fiber are generated by wrapping the fiber around a cylindrical shell **70** having two parts **71** and **72** of the body separated by a slit **73** and moving the two parts relative to one another. Thus, the shell **70** is held fixed at anchors **74**, **75**, **76** and **77** and a free edge of the shell is moved radially inwardly and outwardly by an actuator **12**. This causes the free edge **78** of the shell to move relative to the anchor **77** to cause stretching of the portions of the fiber bridging the slot **73**.

(103) The cylindrical shell supporting the wrapped fiber is stably mounted at the fixed anchors and can be moved at high frequency with low forces at the actuator at the edge of the slot. In this way the required signal can be applied to the fiber at the required amplitude and frequencies without difficulty and effectively. In this implementation the actuator may be mounted only to the free edge **78** of the cylinder at the slit, causing the free edge to move under the influence of the vibrating mass of the actuator **12**.

(104) Alternately, the body of the actuator **12** may be fastened to a support base connecting the stationary supports for example adjacent the support **77** and the portion of the actuator that is displaced is fastened to the free edge **78**, causing vibration without influence of stiffness of the edge, mass of the actuator, or resonances thereof. This vibration stretches and releases tension on the fiber(s), thereby modulating the longitudinal strain.

(105) Turning now to FIGS. **20** to **23** and **25** there is shown a different way to introduce a signal that performs the test function without using mechanical actuators which physically disturb the fiber. This signal used can be any of those described above including envelopes of sine waves as well as complex waveforms.

(106) The concept is to introduce into the far end of the fiber a signal that is representative of either a test tone or signal or a signal representative of a nefarious event directly as an optical signal. There are several ways to accomplish this.

(107) The system uses as the monitoring system **10A**, for functionality of the locating vibration sensor, the arrangement described above as the Distributed Acoustic Sensor (DAS), which acts to measure variations in Rayleigh backscattered signal caused by the event to be detected. That is the receiver **10B** in the monitoring system **10A** is arranged to detect the signals caused by the Rayleigh backscattering which can be analyzed by the signal analysis system **10D** to determine the nature of the reflections and whether they are representative of an event to be monitored and also the timing of the reflections to determine the time of transmission relative to the original pulse and hence the location of the event which caused the reflection.

(108) This backscatter is typically 5 orders of magnitude lower than the incident optical power,

which requires any externally introduced signal to be of similar magnitude. Additionally, the DAS system is sensitive to reflection such as those caused by poorly mated, faulty, or missing connectors.

(109) When introducing a test signal therefore from a test signal source, typically a source laser, a low reflection must be presented to the monitoring equipment. As DAS systems utilize for the monitor signal source very narrow spectrum lasers of tightly controlled wavelength, great care must be taken to select and control the test signal source laser.

(110) As shown in FIG. **20** in a first method there is shown a secondary backscatter method. Rather than injecting a signal directly into the monitored fiber, an optical transmitter source **91** injects the test signal into a length **92** of optical fiber which is connected to the fiber to be monitored through a coupler **90** which can also be a splitter, circulator, or other such device. The test signal passes through the coupler **90**, is launched into the terminated fiber, and is then backscattered into the fiber **10**. The added fiber can be properly terminated in a low reflectance manner.

(111) As shown in FIG. **21** in a second method there is provided a direct injection of a test signal from a laser **94** into far end **95** of the fiber **10**. Angled connectors and low-reflectance optics are used to control any back reflection back into the monitoring device.

(112) As this DAS system measures Rayleigh backscattering, whose signal amplitude is roughly 5 orders of magnitude below incident power, it is important to not reflect signals back to the monitoring system. Any device that is put at the far end must be as non-reflective as possible. In a normal installation, and at the far end of an installation of the device as described above, the fiber is terminated for a very low reflection. A high reflection disturbs the detection system within the monitor.

(113) Additionally, the amplitude should be lowered to correspond to a backscattered signal. This can be achieved by introduction of in-line attenuators **96**.

(114) As shown in FIG. **22** in a third method there is provided a modification of the direct injection method, differentiated by disabling, by an optical shutter **96A** the laser source **94** at the transmit section of the monitoring device while continuing to monitor with the receiver at the monitor device. Alternatively, timing could be such that the transmit signal is sent during off times that are conventional in a pulsed laser device. The expression “pulsed laser” is meant to indicate a laser that varies in amplitude, turns on and off, or one whose optical path is interrupted as a function of time. This allows the monitor system receiver to detect an injected signal in the absence of a monitor signal, and allows more sophisticated messaging in the test signal that is not obscured in the monitor signal.

(115) That is, as the end of fiber reflection is problematic if not tamed, this can be used to advantage as shown in FIG. **23** by controlling it as a signaling mechanism. A low internal reflectance variable attenuator **97**, optical switch **98**, or both are placed inline with a high reflectance termination **99**. This termination can be a simple unterminated or air-gap connector, or it can be a greater reflection such as a reflective deposition on a connector face, such as gold. This reflection can be turned on or off, creating an effective end of line signal that provides both features as described above.

(116) Additionally, the variable attenuator **97** may be modulated such as with a sine wave, causing the end reflection to vary at a pre-determined rate. This can be detected by the signal analysis system **10D** and hence adds an additional layer of security as the monitoring device **10A** will watch for that frequency at that precise location.

(117) As the DAS monitoring system is a time based system wherein transmission time of flight is indicative of distance from the monitoring apparatus, in a preferred embodiment, the test signal source laser **94** in FIG. **21** should be pulsed at a specific time and duration. If this light were to be continuous wave (CW), there would be no indication of location, and the signal obtained by the receiver at the monitoring system **10A** would be treated as a DC offset by the monitoring

equipment. If the light from source laser **94** is pulsed at the required time relative to the monitor pulse, it can be made to appear at any location along the fiber.

(118) In FIG. **21**, a timing signal from the monitor system transmits through an interface **10F** a control signal to a communications interface **10E** at a remote location, typically the far end, and on to the controller, which signals the transmitter **94** when to fire.

(119) Turning now to another alternative arrangement shown in FIG. **25** which is used in a system where no communication is available. In this arrangement, a portion of the monitor signal can be detected using the coupler **16B** and receiver **16C** of the arrangement shown in FIG. **14**. The pulse repetition rate of the monitor signal may be known or measured, and the test signal injected by the transmission laser **94** can be timed accordingly.

(120) For example, in a 10 km installation, one might use a 500 nS monitor pulse from the monitoring system **10A**. This causes a roundtrip where the pulse travels the entire distance and reflections return the entire distance for the 10 km in 100 μ S. In such an installation, the monitor laser would fire typically at a repetition rate of 10 kHz.

(121) For the system utilizing the test signal injected by the laser **94**, for example, firing the inject signal 50 μ S after the monitor laser fires makes the signal appear as though it is located at the end of the fiber. A 500 nS pulse from the laser **94** thus appears like a reflection. Alternatively, the injected pulse from the pulse can be formed into any shape as described above such as an envelope of sinewaves or a complex waveform.

(122) Additionally, this delay of the pulse from the laser **94** relative to the received pulse from the monitor signal can be adjusted to change the apparent location of the virtual disturbance. As the monitor laser is fired in a continuous frequency, a virtual injected signal from laser **94** can fire prior to receipt of the monitor signal, thus acting to locate the apparent disturbance closer to the monitor system end of the fiber.

(123) For example, after an initial monitor laser pulse is detected by the receiver **16C**, where the controller is programmed to know that the monitor signal laser will fire every 100 μ S, an injection from the test signal laser **94** at a time 50 μ S before the expected receipt of all subsequent laser pulses gives the illusion of a disturbance at the midpoint of the fiber. This virtual disturbance, therefore, can therefore be “tuned” to any location along the fiber.

(124) The pertinent specifications of the monitor signal laser need to be reproduced in the test signal injected laser; such as wavelength, spectral width, scattered optical power.

(125) In one embodiment, an optical isolator **100** as shown in FIG. **25** is provided which limits the reflections from the face of the injection laser **94** from launching into the fiber by presenting a low back reflection. As the wavelengths of the monitor signal laser and the injected signal laser are quite close, the isolator **100** keeps incoming monitor laser light passing through the coupler **16B** from reaching the laser **94** which can cause interference and destabilization of the laser **94**.

(126) FIG. **24** is a schematic illustration of the operating components which form the elements of the invention for attachment to the fiber to be monitored where the signal in the fiber is applied by optically injecting the signal from a transmitter **16Y** into the fiber using an optical coupler **16X** which combines the monitor signal with the injected signal. This holds true for closed loop systems which are non-locating in regard to the location of the event as well as systems which act to locate the event along the fiber.

(127) In the closed loop system, the monitoring system transmitter injects a light signal into the fiber, which is typically looped back to the receiver. At the loopback point, an optical coupler enables a test signal to be superimposed on the monitor signal, simulating the fluctuation in signal representative of an intrusion or other disturbance.

Claims

1. A system for facilitating optical fiber monitoring for disturbance events, the system comprising: one or more processors configured to perform operations comprising: performing, via a detection system, monitoring of an optical fiber to detect changes in the optical fiber that are indicative of one or more events to be monitored; during the monitoring of the optical fiber, transmitting, via a light source, a monitor signal along the optical fiber, the monitor signal being received by the detection system as light from the optical fiber; while the detection system continues to receive light from the optical fiber in connection with the transmitting of the monitor signal, periodically causing, via a device at a predetermined location along the optical fiber, a test disturbance event to occur at the predetermined location; and in response to a failure of the detection system to detect test-related changes in the optical fiber caused by the test disturbance event that occurred at the predetermined location, generating a notification indicating an issue with the detection system.
2. The system of claim 1, wherein periodically causing the test disturbance event comprises periodically causing, via the device at the predetermined location along the optical fiber, a test signal to be produced at the predetermined location while the detection system continues to receive light from the optical fiber in connection with the transmitting of the monitor signal, and wherein generating the notification comprises generating the notification indicating the issue with the detection system in response to the failure of the detection system to detect the test-related changes in the optical fiber caused by the test signal produced at the predetermined location.
3. The system of claim 2, wherein periodically causing the test signal to be produced comprises periodically causing, via the device at the predetermined location along the optical fiber, the test signal to be optically injected into the optical fiber at the predetermined location while the detection system continues to receive light from the optical fiber in connection with the transmitting of the monitor signal, and wherein generating the notification comprises generating the notification indicating the issue with the detection system in response to the failure of the detection system to detect the test-related changes in the optical fiber caused by the test signal optically injected into the optical fiber at the predetermined location.
4. The system of claim 1, wherein periodically causing the test disturbance event comprises periodically causing, via the device at the predetermined location along the optical fiber, a vibration to occur at the predetermined location while the detection system continues to receive light from the optical fiber in connection with the transmitting of the monitor signal, wherein generating the notification comprises generating the notification indicating the issue with the detection system in response to the failure of the detection system to detect the test-related changes in the optical fiber caused by the vibration that occurred at the predetermined location.
5. The system of claim 4, wherein periodically causing the vibration comprises periodically causing, via an actuator at the predetermined location along the optical fiber, the vibration to occur at the predetermined location while the detection system continues to receive light from the optical fiber in connection with the transmitting of the monitor signal, wherein generating the notification comprises generating the notification indicating the issue with the detection system in response to the failure of the detection system to detect the test-related changes in the optical fiber caused by the vibration that occurred at the predetermined location.
6. A system comprising: one or more processors configured to perform operations comprising: performing, via a detection system, monitoring of a data transmission cable to detect one or more events; transmitting a monitor signal along the data transmission cable; while the detection system continues to receive a signal from the data transmission cable, causing, via one or more devices along the data transmission cable, a test event to occur along the data transmission cable; and generating a notification indicating an issue with the detection system based on a determination that the detection system failed to detect one or more changes in the data transmission cable caused by the test event that occurred along the data transmission cable.
7. The system of claim 6, wherein causing the test event comprises causing, via the one or more

devices along the data transmission cable, a test signal to be produced along the data transmission cable while the detection system continues to receive a signal from the data transmission cable, and wherein generating the notification comprises generating the notification indicating the issue with the detection system based on the determination that the detection system failed to detect the one or more changes in the data transmission cable caused by the test signal produced along the data transmission cable.

8. The system of claim 7, wherein causing the test signal to be produced comprises causing, via the one or more devices along the data transmission cable, the test signal to be optically injected into the data transmission cable along the data transmission cable while the detection system continues to receive a signal from the data transmission cable, and wherein generating the notification comprises generating the notification indicating the issue with the detection system based on the determination that the detection system failed to detect the one or more changes in the data transmission cable caused by the test signal optically injected into the data transmission cable.

9. The system of claim 6, wherein causing the test event comprises causing, via the one or more devices along the data transmission cable, a vibration to occur along the data transmission cable while the detection system continues to receive a signal from the data transmission cable, wherein generating the notification comprises generating the notification indicating the issue with the detection system based on the determination that the detection system failed to detect the one or more changes in the data transmission cable caused by the vibration that occurred along the data transmission cable.

10. The system of claim 9, wherein causing the vibration comprises causing, via an actuator along the data transmission cable along the data transmission cable, the vibration to occur along the data transmission cable while the detection system continues to receive a signal from the data transmission cable, wherein generating the notification comprises generating the notification indicating the issue with the detection system based on the determination that the detection system failed to detect the one or more changes in the data transmission cable caused by the vibration that occurred along the data transmission cable.

11. A method comprising: monitoring, via a detection system, a data transmission cable to detect one or more events; transmitting a monitor signal along the data transmission cable; while the detection system continues to receive a signal from the data transmission cable, causing, via one or more devices along the data transmission cable, a test event to occur along the data transmission cable; determining that the detection system failed to detect one or more changes in the data transmission cable caused by the test event that occurred along the data transmission cable; and generating a notification indicating an issue with the detection system based on the determination that the detection system failed to detect the one or more changes.

12. The method of claim 11, wherein causing the test event to occur along the data transmission cable comprises periodically causing, via the one or more devices along the data transmission cable, the test event to occur along the data transmission cable while the detection system continues to receive a signal from the data transmission cable.

13. The method of claim 11, wherein causing the test event to occur along the data transmission cable comprises causing, via the one or more devices along the data transmission cable, at one or more scheduled times, the test event to occur along the data transmission cable while the detection system continues to receive a signal from the data transmission cable.

14. The method of claim 11, wherein causing the test event to occur along the data transmission cable comprises causing, via the one or more devices along the data transmission cable, at one or more random times initiated by the one or more devices, the test event to occur along the data transmission cable while the detection system continues to receive a signal from the data transmission cable.

15. The method of claim 11, wherein causing the test event comprises causing, via the one or more devices along the data transmission cable, a test signal to be produced along the data transmission

cable while the detection system continues to receive a signal from the data transmission cable, and wherein generating the notification comprises generating the notification indicating the issue with the detection system based on the determination that the detection system failed to detect the one or more changes in the data transmission cable caused by the test signal produced along the data transmission cable.

16. The method of claim 15, wherein causing the test signal to be produced comprises causing, via the one or more devices along the data transmission cable, the test signal to be optically injected into the data transmission cable while the detection system continues to receive a signal from the data transmission cable, and wherein generating the notification comprises generating the notification indicating the issue with the detection system based on the determination that the detection system failed to detect the one or more changes in the data transmission cable caused by the test signal optically injected into the data transmission cable.

17. The method of claim 11, wherein causing the test event comprises causing, via the one or more devices along the data transmission cable, a vibration to occur along the data transmission cable, wherein generating the notification comprises generating the notification indicating the issue with the detection system based on the determination that the detection system failed to detect the one or more changes in the data transmission cable caused by the vibration that occurred along the data transmission cable.

18. The method of claim 17, wherein causing the vibration comprises causing, via an actuator along the data transmission cable, the vibration to occur along the data transmission cable, wherein generating the notification comprises generating the notification indicating the issue with the detection system based on the determination that the detection system failed to detect the one or more changes in the data transmission cable caused by the vibration that occurred along the data transmission cable.

19. The method of claim 11, wherein causing the test event comprises causing, via the one or more devices along the data transmission cable, a displacement of the data transmission cable to occur, wherein generating the notification comprises generating the notification indicating the issue with the detection system based on the determination that the detection system failed to detect the one or more changes in the data transmission cable caused by the displacement.

20. The method of claim 11, wherein causing the test event comprises controlling the one or more devices along the data transmission cable to heat or cool a portion of the data transmission cable, wherein generating the notification comprises generating the notification indicating the issue with the detection system based on the determination that the detection system failed to detect the one or more changes in the data transmission cable caused by the heating or cooling of the portion of the data transmission cable.
